MATH1101: Differential Calculus & Analytical Geometry

Complete Exam Notes · 1st Year 1st Semester · PUST · Module 3 (Units 6, 7, 8 & 9): Mean Value Theorems, Series, L'Hôpital's Rule & Extrema

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Module 3 Overview

Units Covered in Module 3:

- **Unit 6:** Rolle's Theorem and Lagrange's Mean Value Theorem (Statements, Geometric Interpretations, Proofs, Class Examples & All 17 Handout Problems).
- **Unit 7:** Power Series, Taylor's Series, and Maclaurin's Series (Derivations & Anton Homework Sets 9.8 and 9.9 Ex. 17).
- **Unit 8:** Indeterminate Forms and L'Hôpital's Rule (Class Problems, PUST Past Exam Questions & Anton Homework Set 3.6).
- **Unit 9:** Increasing/Decreasing Functions, Concavity, Inflection Points, Maxima & Minima, and Curve Tracing.

Master Roadmap (MATH 1101)

Mod.	Units & Core Syllabus Topics	Main Sources	Status
1	Units 1–3: Functions, Domain/Range, Graphs, Inverse, Limits & Continuity	Class Notes · Anton Ch. 0, 1 · PUST 20–23	Module 1
2	Units 4–5: Differentiability, Geometric Meaning, Extrema, y_n & Leibniz	Class Notes · Anton Ch. 2, 4 · PUST 20–23	Module 2
3	Units 6–9: Rolle's & MVT, Taylor/Maclaurin, L'Hôpital & Curve Tracing	Class Notes · Anton Ch. 3, 4, 9 · Handout	✓ **This file**
4	Units 10–13: Tangents, Asymptotes, Curvature, Partial Deriv., Euler & Jacobian	Akib Notes · Anton Ch. 13 · PUST 20–23	Module 4
5	Units 14–15: 2D Coordinates, Transformation of Axes & Pair of Straight Lines	Anton Ch. 10 · Rahman & Bhattacharjee	Module 5
6	Units 16–18: Conic Reduction, Parabola, Ellipse, Hyperbola & System of Circles	Anton Ch. 10 · Rahman & Bhattacharjee	Module 6
7	Units 19–21: 3D Coordinates, DCs/DRs, Projections, Planes & Skew Lines (S.D.)	Anton Ch. 11 · Rahman & Bhattacharjee	Module 7
8	Units 22–24: Sphere, Cone, Right Circular Cylinder & 3D Central Conicoids	Anton Ch. 11 · Rahman & Bhattacharjee	Module 8

Contents

1 Unit 6: Rolle's Theorem and Lagrange's Mean Value Theorem **2**

1.1 Rolle's Theorem: Statement, Geometric Meaning & Proof . . . 2

1.2 Lagrange's Mean Value Theorem (MVT) . . . 3

1.3 Solved Problems from the Rolle's & MVT Handout (Problems 1–17) . . . 4

2 Unit 7: Power Series, Taylor's Series, and Maclaurin's Series **5**

2.1 Definitions of Power Series, Taylor's Series & Maclaurin's Series . . . 5

2.2 Standard Maclaurin Series Expansions . . . 5

2.3 Solved Anton Homework Sets 9.8 (Ex. 1–22, 29–50) & 9.9 (Ex. 17) . . . 6

3 Unit 8: Indeterminate Forms and L'Hôpital's Rule **7**
3.1 The Seven Indeterminate Forms & Reduction Rules . . . 7
3.2 Solved Class Lecture & PUST Past Exam Problems on L'Hôpital's Rule . . . 7
3.3 Solved Anton Homework Set 3.6 (Ex. 7–45, 59–62, 66) . . . 8

4 Unit 9: Extrema, Concavity, Inflection Points & Curve Tracing **8**
4.1 Definitions and Derivative Tests . . . 8
4.2 Working Rule for Curve Tracing . . . 10
4.3 Past-Question Map for Module 3 . . . 10

1 Unit 6: Rolle's Theorem and Lagrange's Mean Value Theorem

1.1 Rolle's Theorem: Statement, Geometric Meaning & Proof

Rolle's Theorem (Class Lecture & PUST EECE 2022–23 Exam Q2a) [5.5 Marks]
Statement: Let $f(x)$ be a real-valued function defined on the closed interval $[a, b]$ such that:

1. $f(x)$ is **continuous** on the closed interval $[a, b]$,
2. $f(x)$ is **differentiable** on the open interval (a, b), and
3. $f(a) = f(b)$.

Then there exists at least one real number $c \in (a, b)$ such that:

$$f'(c) = 0.$$

Geometric Representation: If a continuous, smooth curve starts at $(a, f(a))$ and ends at the same height $(b, f(b))$, then there is at least one interior point $P(c, f(c))$ with $a < c < b$ where the tangent line to the curve is parallel to the x-axis (slope $f'(c) = 0$).

Proof (Exam Style): Since $f(x)$ is continuous on $[a, b]$, it attains an absolute maximum M and an absolute minimum m on $[a, b]$.

- **Case 1 ($M = m$):** Then $f(x)$ is constant on $[a, b]$, so $f'(c) = 0$ for every $c \in (a, b)$.
- **Case 2 ($M > m$):** Since $f(a) = f(b)$, at least one of M or m differs from $f(a)$ and must occur at some interior point $c \in (a, b)$. Since $f(x)$ has a local extremum at c and is differentiable at c, $f'(c) = 0$. ■

Problem 6.1 (Class Lecture): Verify Rolle's Theorem for $f(x) = x^2 - 4x + 3$ on $[1, 3]$.

Solution (Exam Style):

1. Being a polynomial, $f(x)$ is continuous on $[1, 3]$.
2. $f'(x) = 2x - 4$ exists finitely on $(1, 3)$, so $f(x)$ is differentiable on $(1, 3)$.
3. Checking endpoint values:

   $$f(1) = 1^2 - 4(1) + 3 = 0$$
   $$f(3) = 3^2 - 4(3) + 3 = 0 \implies f(1) = f(3) = 0.$$

   All three conditions of Rolle's Theorem are satisfied. Now setting $f'(c) = 0$:

   $$2c - 4 = 0$$
   $$\implies c = 2 \in (1, 3).$$

   Hence Rolle's Theorem is verified.

Problem 6.2 (PUST EECE 2021–22 Exam Q1c [4 Marks]): Verify Rolle's Theorem for $f(x) = x(x-3)^2$ on $[0,3]$.

Solution (Exam Style):

$f(x) = x(x-3)^2 = x^3 - 6x^2 + 9x$ [Continuous on $[0,3]$]
$f'(x) = 3x^2 - 12x + 9$ [Differentiable on $(0,3)$]
$f(0) = 0, \qquad f(3) = 3(0)^2 = 0 \implies f(0) = f(3) = 0.$

Setting $f'(c) = 0$:

$$\begin{aligned} &3c^2 - 12c + 9 = 0 \\ &\implies 3(c-1)(c-3) = 0 \\ &\implies c = 1,\ 3. \end{aligned}$$

Since $c = 1 \in (0,3)$, Rolle's Theorem is verified.

1.2 Lagrange's Mean Value Theorem (MVT)

Lagrange's Mean Value Theorem (Class Lecture & PUST EECE 2021–22 Q1b) [5 Marks]
Statement: If a function $f(x)$ is:

1. **continuous** on the closed interval $[a,b]$, and
2. **differentiable** on the open interval (a,b),

then there exists at least one real number $c \in (a,b)$ such that:

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Geometric Representation: There is at least one point $P(c, f(c))$ on the curve between $A(a, f(a))$ and $B(b, f(b))$ where the tangent line is parallel to the secant chord AB.

Proof (Exam Style): Define the auxiliary function $\phi(x) = f(x) + Ax$, where the constant A is chosen so that $\phi(a) = \phi(b)$:

$$\begin{aligned} &f(a) + Aa = f(b) + Ab \\ &\implies A(b-a) = -\big[f(b) - f(a)\big] \\ &\implies A = -\frac{f(b) - f(a)}{b - a}. \end{aligned}$$

Since $\phi(x)$ is continuous on $[a,b]$, differentiable on (a,b), and $\phi(a) = \phi(b)$, by Rolle's Theorem there exists $c \in (a,b)$ such that $\phi'(c) = 0$:

$$\begin{aligned} &f'(c) + A = 0 \\ &\implies f'(c) = -A = \frac{f(b) - f(a)}{b - a}. \quad \blacksquare \end{aligned}$$

Problem 6.3 (Class Lecture): Verify Lagrange's MVT for $f(x) = x^3 - 4x$ on $[-2,1]$.

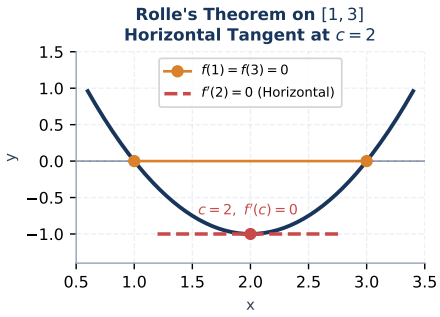
Solution (Exam Style):

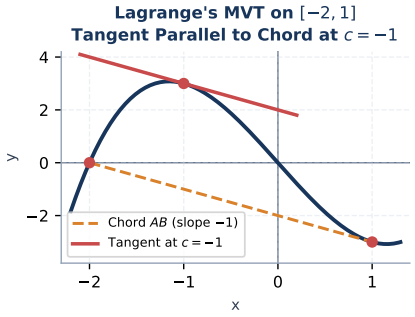
$f(x) = x^3 - 4x$ is continuous on $[-2,1]$ and $f'(x) = 3x^2 - 4$ exists on $(-2,1)$,
$f(-2) = (-2)^3 - 4(-2) = 0, \qquad f(1) = 1^3 - 4(1) = -3.$

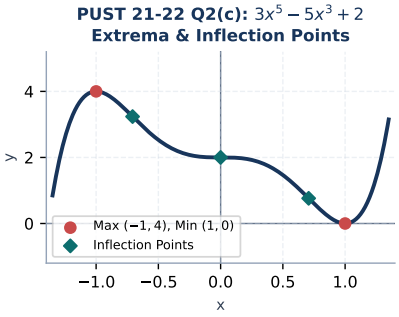
By Lagrange's Mean Value Theorem:

$$f'(c) = \frac{f(1) - f(-2)}{1 - (-2)}$$

$$\implies 3c^2 - 4 = \frac{-3-0}{3} = -1$$
$$\implies 3c^2 = 3 \implies c = \pm 1$$
$$\therefore c = -1 \in (-2, 1) \quad \text{[Hence MVT is verified]}.$$







1.3 Solved Problems from the Rolle's & MVT Handout (Problems 1–17)

1. **Handout #2:** Verify Rolle's Theorem for $f(x) = \ln\left[\dfrac{x^2+ab}{x(a+b)}\right]$ on $[a,b]$ $(0 \notin [a,b])$.

$$f(x) = \ln(x^2+ab) - \ln x - \ln(a+b)$$
$$f(a) = \ln\left[\frac{a(a+b)}{a(a+b)}\right] = \ln 1 = 0, \qquad f(b) = \ln\left[\frac{b(b+a)}{b(a+b)}\right] = \ln 1 = 0$$
$$f'(c) = \frac{2c}{c^2+ab} - \frac{1}{c} = \frac{c^2-ab}{c(c^2+ab)} = 0$$
$$\implies c^2 = ab \implies c = \sqrt{ab} \in (a,b) \quad \text{[Geometric mean lies between } a \text{ and } b\text{]}.$$

2. **Handout #3 & PUST EECE 2020–21 Exam Q2(b) [4.5 Marks]:** Verify Lagrange's MVT for $f(x) = x(x-1)(x-2)$ on $\left[0, \frac{1}{2}\right]$.

$$f(x) = x^3 - 3x^2 + 2x, \qquad f'(x) = 3x^2 - 6x + 2$$
$$f(0) = 0, \qquad f\left(\frac{1}{2}\right) = \frac{1}{2}\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right) = \frac{3}{8}$$
$$f'(c) = \frac{f(1/2) - f(0)}{1/2 - 0}$$
$$\implies 3c^2 - 6c + 2 = \frac{3/8}{1/2} = \frac{3}{4}$$
$$\implies 12c^2 - 24c + 5 = 0$$
$$\implies c = \frac{24 \pm \sqrt{576-240}}{24} = \frac{6 \pm \sqrt{21}}{6}$$
$$\therefore c = \frac{6-\sqrt{21}}{6} \approx 0.236 \in \left(0, \frac{1}{2}\right).$$

3. **Handout #5, 6, 13, 17 (Exponential-Trigonometric Rolle's Theorem Problems):**

#5: $f(x) = e^x \sin x$ on $[0, \pi]$, $\quad f(0) = f(\pi) = 0$
$$f'(c) = e^c(\sin c + \cos c) = 0 \implies \tan c = -1 \quad [\because e^c \neq 0]$$
$$\implies c = \frac{3\pi}{4} \in (0, \pi).$$

#6: $f(x) = e^{2x}(\sin 2x - \cos 2x)$ on $\left[\dfrac{\pi}{8}, \dfrac{5\pi}{8}\right]$, $\quad f\left(\dfrac{\pi}{8}\right) = f\left(\dfrac{5\pi}{8}\right) = 0$
$$f'(c) = 4e^{2c}\sin 2c = 0 \implies 2c = \pi \implies c = \frac{\pi}{2} \in \left(\frac{\pi}{8}, \frac{5\pi}{8}\right).$$

#13: $f(x) = e^x(\sin x - \cos x)$ on $\left[\frac{\pi}{4}, \frac{5\pi}{4}\right] \implies f'(c) = 2e^c \sin c = 0 \implies c = \pi \in \left(\frac{\pi}{4}, \frac{5\pi}{4}\right)$.

#17: $f(x) = e^{-x}\sin x$ on $[0, \pi] \implies f'(c) = e^{-c}(\cos c - \sin c) = 0 \implies \tan c = 1 \implies c = \frac{\pi}{4} \in (0, \pi)$.

4. **Handout #9:** Rolle's Theorem holds for $f(x) = x^3 + ax^2 + bx$ on $[1, 2]$ at $c = \frac{4}{3}$. Find a and b.

$$f(1) = f(2) \implies 1 + a + b = 8 + 4a + 2b \implies 3a + b = -7$$
$$f'\left(\frac{4}{3}\right) = 0 \implies 3\left(\frac{16}{9}\right) + 2a\left(\frac{4}{3}\right) + b = 0 \implies 8a + 3b = -16$$
$$\therefore a = -5, \qquad b = 8.$$

*(Remaining Handout Answers: **#1:** $c = \frac{1}{7}$; **#4:** $c = 2$, point $(2, -7)$; **#7:** $c = \frac{7}{2}$; **#8:** $c = \frac{9}{4}$, point $(\frac{9}{4}, \frac{1}{2})$; **#10:** $c = \frac{\pi}{4}$, point $(\frac{\pi}{4}, \sqrt{2} - 1)$; **#11:** $\cos c = \frac{1 \pm \sqrt{33}}{8}$; **#12:** $4c^2 - 4c - 3 = 0 \implies c = \frac{3}{2} \in (1, 4)$; **#14:** $2\cos^2 c + \cos c - 1 = 0 \implies c = \frac{\pi}{3}$; **#15:** $c = 0$, point $(0, \ln\frac{2}{3})$; **#16:** $-\ln c = 1 - 2\ln 2 \implies c = \frac{4}{e} \in (1, 2)$.)*

2 Unit 7: Power Series, Taylor's Series, and Maclaurin's Series

2.1 Definitions of Power Series, Taylor's Series & Maclaurin's Series

1. **Power Series:** $\sum_{n=0}^{\infty} c_n x^n$ (centered at 0) or $\sum_{n=0}^{\infty} c_n (x-a)^n$ (centered at a).
2. **Taylor's Series about $x = a$:**

$$f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \frac{(x-a)^3}{3!}f'''(a) + \cdots = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n.$$

3. **Maclaurin's Series (Taylor's Series at $a = 0$):**

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \cdots = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n.$$

2.2 Standard Maclaurin Series Expansions

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \qquad x \in (-\infty, \infty)$$
$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{(2k+1)!}, \qquad x \in (-\infty, \infty)$$
$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{(2k)!}, \qquad x \in (-\infty, \infty)$$
$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots = \sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^n}{n}, \qquad x \in (-1, 1]$$
$$(1+x)^m = 1 + mx + \frac{m(m-1)}{2!}x^2 + \frac{m(m-1)(m-2)}{3!}x^3 + \cdots, \qquad |x| < 1.$$

PUST EECE 2021–22 Exam Q4(a) [4.5 Marks] & 2022–23 Exam Q4(a) [4 Marks]

Problem A (2021–22 Q4a): State Maclaurin's theorem. Expand $\ln(1+x)$ in an infinite series.
Solution (Exam Style):

$$f(x)=\ln(1+x) \implies f(0)=\ln 1=0$$
$$f'(x)=(1+x)^{-1} \implies f'(0)=1$$
$$f''(x)=-(1+x)^{-2} \implies f''(0)=-1$$
$$f'''(x)=2!(1+x)^{-3} \implies f'''(0)=2!$$
$$f^{(iv)}(x)=-3!(1+x)^{-4} \implies f^{(iv)}(0)=-3!$$

Substituting into $f(x)=f(0)+xf'(0)+\frac{x^2}{2!}f''(0)+\frac{x^3}{3!}f'''(0)+\frac{x^4}{4!}f^{(iv)}(0)+\cdots$:

$$\ln(1+x)=x-\frac{x^2}{2}+\frac{x^3}{3}-\frac{x^4}{4}+\cdots+(-1)^{n-1}\frac{x^n}{n}+\cdots.$$

Problem B (2022–23 Q4a): Expand $y=e^x\sec x$ in ascending powers of x up to the x^3 term using Maclaurin's series.
Solution (Exam Style):

$$y=e^x\sec x \implies y(0)=1$$
$$y_1=e^x\sec x+e^x\sec x\tan x=y(1+\tan x) \implies y_1(0)=1(1+0)=1$$
$$y_2=y_1(1+\tan x)+y\sec^2 x \implies y_2(0)=1(1)+1(1)=2$$
$$y_3=y_2(1+\tan x)+2y_1\sec^2 x+2y\sec^2 x\tan x \implies y_3(0)=2(1)+2(1)(1)+0=4$$

Substituting into Maclaurin's series:

$$e^x\sec x=y(0)+xy_1(0)+\frac{x^2}{2!}y_2(0)+\frac{x^3}{3!}y_3(0)+\cdots$$
$$=1+x+x^2+\frac{2}{3}x^3+\cdots.$$

2.3 Solved Anton Homework Sets 9.8 (Ex. 1–22, 29–50) & 9.9 (Ex. 17)

1. **Anton 9.8 Ex. 1–18 (Maclaurin & Taylor Series): 1.** $e^{-x}=\sum_{k=0}^{\infty}\frac{(-1)^k x^k}{k!}$; **2.** $e^{ax}=\sum_{k=0}^{\infty}\frac{a^k x^k}{k!}$; **3.** $\cos\pi x=\sum_{k=0}^{\infty}\frac{(-1)^k\pi^{2k}x^{2k}}{(2k)!}$; **4.** $\sin\pi x=\sum_{k=0}^{\infty}\frac{(-1)^k\pi^{2k+1}x^{2k+1}}{(2k+1)!}$; **5.** $\ln(1+x)=\sum_{k=1}^{\infty}\frac{(-1)^{k-1}x^k}{k}$; **6.** $\frac{1}{1+x}=\sum_{k=0}^{\infty}(-1)^k x^k$; **7.** $\cosh x=\sum_{k=0}^{\infty}\frac{x^{2k}}{(2k)!}$; **8.** $\sinh x=\sum_{k=0}^{\infty}\frac{x^{2k+1}}{(2k+1)!}$; **9.** $x\sin x=\sum_{k=0}^{\infty}\frac{(-1)^k x^{2k+2}}{(2k+1)!}$; **10.** $xe^x=\sum_{k=0}^{\infty}\frac{x^{k+1}}{k!}$; **11.** $\sum_{k=0}^{\infty}\frac{e(x-1)^k}{k!}$; **12.** $\sum_{k=0}^{\infty}\frac{(-1)^k(x-\ln 2)^k}{2\cdot k!}$; **13.** $\sum_{k=0}^{\infty}-(x+1)^k$; **14.** $\sum_{k=0}^{\infty}\frac{(-1)^k(x-3)^k}{5^{k+1}}$; **15.** $\sum_{k=0}^{\infty}\frac{(-1)^k\pi^{2k}(x-1/2)^{2k}}{(2k)!}$; **16.** $\sum_{k=0}^{\infty}\frac{(-1)^{k+1}(x-\pi/2)^{2k+1}}{(2k+1)!}$; **17.** $\sum_{k=1}^{\infty}\frac{(-1)^{k-1}(x-1)^k}{k}$; **18.** $1+\sum_{k=1}^{\infty}\frac{(-1)^{k-1}(x-e)^k}{ke^k}$.
2. **Anton 9.8 Ex. 19–22 & 29–50 (Radius R and Interval I of Convergence):**

Ex. 45: $u_k=\dfrac{(-1)^{k+1}(x+1)^k}{k}$

$$\lim_{k\to\infty}\left|\frac{u_{k+1}}{u_k}\right|=\lim_{k\to\infty}\frac{k}{k+1}|x+1|=|x+1|<1 \quad \text{[Ratio Test]}$$
$$\implies R=1, \quad -2<x<0, \quad \text{and checking endpoints gives } I=(-2,0].$$

*(Complete answers for 19–22 & 29–50: **19.** $\frac{1}{1+x},(-1,1)$; **20.** $\frac{1}{1-x^2},(-1,1)$; **21.** $\frac{1}{3-x},(1,3)$; **22.** $\frac{1}{x+4},(-4,-2)$; **29.** $R=1,[-1,1)$; **30.** $R=\frac{1}{3},(-\frac{1}{3},\frac{1}{3})$; **31.** $R=\infty,\mathbb{R}$; **32.** $R=0,\{0\}$; **33.** $R=\frac{1}{5},[-\frac{1}{5},\frac{1}{5}]$; **34.** $R=1,[-1,1)$; **35.** $R=1,[-1,1]$; **36.** $R=\frac{1}{2},(-\frac{1}{2},\frac{1}{2}]$; **37.** $R=1,(-1,1]$; **38–39.** $R=\infty,\mathbb{R}$; **40–41.** $R=1,[-1,1]$; **42.** $R=\infty,\mathbb{R}$; **43.** $R=\frac{4}{3},(-\frac{19}{3},-\frac{11}{3})$; **44.** $R=2,(1,5)$; **46.** $R=1,[3,5]$; **47.** $R=1,[-2,0]$; **48.** $R=0,\{2\}$; **49.** $R=\infty,\mathbb{R}$; **50.** $R=8,(-\frac{13}{2},\frac{19}{2})$.)*

3. **Anton 9.9 Ex. 17 (Binomial Maclaurin Series):**

(a) $(1+x)^{-1} = 1 - x + x^2 - x^3 + \cdots = \sum_{k=0}^{\infty} (-1)^k x^k,$

(b) $(1+x)^{1/3} = 1 + \frac{1}{3}x - \frac{1}{9}x^2 + \frac{5}{81}x^3 - \cdots,$

(c) $(1+x)^{-3} = 1 - 3x + 6x^2 - 10x^3 + \cdots = \sum_{k=0}^{\infty} (-1)^k \frac{(k+1)(k+2)}{2} x^k.$

3 Unit 8: Indeterminate Forms and L'Hôpital's Rule

3.1 The Seven Indeterminate Forms & Reduction Rules

The seven indeterminate forms are:

$$\frac{0}{0}, \quad \frac{\infty}{\infty}, \quad 0 \cdot \infty, \quad \infty - \infty, \quad 0^0, \quad 1^\infty, \quad \infty^0.$$

Important Note: Standard Reductions to $\frac{0}{0}$ or $\frac{\infty}{\infty}$

1. **Form $0 \cdot \infty$:** Rewrite $f(x) \cdot g(x)$ as $\dfrac{f(x)}{1/g(x)}$ or $\dfrac{g(x)}{1/f(x)}$.
2. **Form $\infty - \infty$:** Combine fractions over a common denominator or rationalize.
3. **Exponential Forms $0^0, 1^\infty, \infty^0$:** Let $y = [f(x)]^{g(x)}$, take $\ln y = g(x) \ln f(x) = \dfrac{\ln f(x)}{1/g(x)}$, find $L = \lim \ln y$, and conclude $\lim y = e^L$.

3.2 Solved Class Lecture & PUST Past Exam Problems on L'Hôpital's Rule

PUST EECE 2020–21 Exam Q2(c) [4.5 Marks] & Class Lecture

Evaluate: (i) $\lim_{x\to 0}\left(\frac{1}{x} - \frac{1}{\sin x}\right)$ and (ii) $\lim_{x\to 0}(1+\sin x)^{1/x}$.

Solution (Exam Style):

(i) $L = \lim_{x\to 0}\left(\frac{1}{x} - \frac{1}{\sin x}\right)$ [Form $\infty - \infty$]

$= \lim_{x\to 0} \frac{\sin x - x}{x \sin x}$ [Form $\frac{0}{0}$, applying L'Hôpital's Rule]

$= \lim_{x\to 0} \frac{\cos x - 1}{\sin x + x\cos x}$ [Form $\frac{0}{0}$, applying L'Hôpital's Rule again]

$= \lim_{x\to 0} \frac{-\sin x}{2\cos x - x\sin x} = \frac{0}{2-0} = 0.$

(ii) Let $y = (1+\sin x)^{1/x}$ [Form 1^∞]

$\implies \ln y = \dfrac{\ln(1+\sin x)}{x}$

$\implies \lim_{x\to 0} \ln y = \lim_{x\to 0} \dfrac{\ln(1+\sin x)}{x}$ [Form $\frac{0}{0}$, applying L'Hôpital's Rule]

$= \lim_{x\to 0} \dfrac{\frac{\cos x}{1+\sin x}}{1} = 1$

$\therefore \lim_{x\to 0} y = e^1 = e.$

PUST EECE 2022–23 Exam Q4(b) [5 Marks] & 2021–22 Exam Q4(b) [5 Marks]

Evaluate: (i) $\lim_{x\to0}\dfrac{e^x+e^{-x}-2\cos x}{x\sin x}$, (ii) $\lim_{x\to0}(\cos x)^{\cot^2 x}$, (iii) $\lim_{x\to0}\dfrac{\tan x-x}{x^2\tan x}$, (iv) $\lim_{x\to\pi/2}(\sin x)^{\tan x}$.

Solution (Exam Style):

(i) $L=\lim\limits_{x\to0}\dfrac{e^x+e^{-x}-2\cos x}{x\sin x}$ [Form $\frac{0}{0}$]

$=\lim\limits_{x\to0}\dfrac{e^x-e^{-x}+2\sin x}{\sin x+x\cos x}$ [Form $\frac{0}{0}$]

$=\lim\limits_{x\to0}\dfrac{e^x+e^{-x}+2\cos x}{2\cos x-x\sin x}=\dfrac{1+1+2}{2-0}=2.$

(ii) Let $y=(\cos x)^{\cot^2 x}\implies \ln y=\dfrac{\ln(\cos x)}{\tan^2 x}$ [Form $\frac{0}{0}$]

$\implies \lim\limits_{x\to0}\ln y=\lim\limits_{x\to0}\dfrac{-\tan x}{2\tan x\sec^2 x}=\lim\limits_{x\to0}\dfrac{-1}{2\sec^2 x}=-\dfrac{1}{2}$

$\therefore \lim\limits_{x\to0}y=e^{-1/2}=\dfrac{1}{\sqrt{e}}.$

(iii) $L=\lim\limits_{x\to0}\left(\dfrac{\tan x-x}{x^3}\cdot\dfrac{x}{\tan x}\right)$

$=\lim\limits_{x\to0}\dfrac{\sec^2 x-1}{3x^2}\cdot1=\dfrac{1}{3}\lim\limits_{x\to0}\left(\dfrac{\tan x}{x}\right)^2=\dfrac{1}{3}.$

(iv) Let $y=(\sin x)^{\tan x}\implies \ln y=\dfrac{\ln(\sin x)}{\cot x}$

$\implies \lim\limits_{x\to\pi/2}\ln y=\lim\limits_{x\to\pi/2}\dfrac{\cot x}{-\csc^2 x}=\lim\limits_{x\to\pi/2}(-\sin x\cos x)=0\implies \lim\limits_{x\to\pi/2}y=e^0=1.$

3.3 Solved Anton Homework Set 3.6 (Ex. 7–45, 59–62, 66)

1. **Ex. 29, 31, 36, 66 (Representative Vertical Solutions):**

Ex. 29: $y=(e^x+x)^{1/x}\implies \lim\limits_{x\to0}\ln y=\lim\limits_{x\to0}\dfrac{\ln(e^x+x)}{x}=\lim\limits_{x\to0}\dfrac{e^x+1}{e^x+x}=2\implies \lim\limits_{x\to0}y=e^2.$

Ex. 36: $L=\lim\limits_{x\to0}\left(\dfrac{1}{x}-\dfrac{1}{e^x-1}\right)=\lim\limits_{x\to0}\dfrac{e^x-1-x}{x(e^x-1)}$

$=\lim\limits_{x\to0}\dfrac{e^x-1}{xe^x+e^x-1}=\lim\limits_{x\to0}\dfrac{e^x}{xe^x+2e^x}=\dfrac{1}{2}.$

Ex. 66: Given $\lim\limits_{x\to0}\dfrac{k+\cos lx}{x^2}=-4$

Since denominator $x^2\to0$, $\quad k+\cos0=0\implies k=-1$ [Necessary for finite limit]

$\implies \lim\limits_{x\to0}\dfrac{-1+\cos lx}{x^2}=\lim\limits_{x\to0}\dfrac{-l\sin lx}{2x}=-\dfrac{l^2}{2}=-4$

$\implies l^2=8\implies l=\pm2\sqrt{2}.$

*(Complete Answer Key for Set 3.6: **7.** 1; **8.** $\frac{2}{5}$; **9.** 1; **10.** -1; **11.** -1; **12.** $+\infty$; **13.** 0; **14.** $+\infty$; **15.** $-\infty$; **16.** 0; **17.** 0; **18.** 1; **19.** 2; **20.** $\frac{1}{3}$; **21.** 0; **22.** -2; **23.** π; **24.** 0; **25.** $-\frac{5}{3}$; **26.** 1; **27.** e^{-3}; **28.** e^{-6}; **30.** e^{ab}; **31.** $e^{2/\pi}$; **32.** e^{-2}; **33.** 0; **34.** $\frac{9}{2}$; **35.** $\frac{1}{2}$; **37.** $+\infty$; **38.** 0; **39–45.** 1; **59.** 1; **60.** $\frac{2}{3}$; **61.** DNE; **62.** 0.)*

4 Unit 9: Extrema, Concavity, Inflection Points & Curve Tracing

4.1 Definitions and Derivative Tests

1. **Critical & Stationary Points:** $x=c$ is a critical point if $f'(c)=0$ (stationary point) or $f'(c)$ is undefined.
2. **Monotonicity:** $f'(x)>0\implies$ strictly increasing; $\quad f'(x)<0\implies$ strictly decreasing.

3. **First Derivative Test at $x = c$:** Sign of $f'(x)$ changes $+ \to -$ $\implies$ Local Maximum; changes $- \to +$ $\implies$ Local Minimum; no sign change $\implies$ neither.
4. **Second Derivative Test at $f'(c) = 0$:** $f''(c) < 0$ $\implies$ Local Maximum; $f''(c) > 0$ $\implies$ Local Minimum; $f''(c) = 0$ $\implies$ Test fails (use First Derivative Test).
5. **Concavity & Inflection Point:** $f''(x) > 0$ $\implies$ Concave Up ($\cup$); $f''(x) < 0$ $\implies$ Concave Down ($\cap$). A point where $f''(x)$ changes sign is an **Inflection Point**.

Class Lecture & PUST EECE 2021–22 Exam Q2(c) [4 Marks]

Question: Find the critical points, inflection points, relative extrema, and intervals on which $f(x) = 3x^5 - 5x^3 + 2$ is increasing, decreasing, concave up, and concave down.

Solution (Exam Style):

$f(x) = 3x^5 - 5x^3 + 2$

$f'(x) = 15x^4 - 15x^2 = 15x^2(x-1)(x+1)$

For critical points, $f'(x) = 0 \implies x = -1,\ 0,\ 1.$

Interval	$15x^2$	$(x+1)$	$(x-1)$	**Sign of $f'(x)$**	**Behavior of $f(x)$**
$(-\infty, -1)$	$+$	$-$	$-$	$+$	Increasing
$(-1, 0)$	$+$	$+$	$-$	$-$	Decreasing
$(0, 1)$	$+$	$+$	$-$	$-$	Decreasing
$(1, \infty)$	$+$	$+$	$+$	$+$	Increasing

- **Increasing on:** $(-\infty, -1] \cup [1, \infty)$; **Decreasing on:** $[-1, 1]$.
- **Local Extrema:**

  At $x = -1$: $f_{\max} = f(-1) = 3(-1) - 5(-1) + 2 = 4$ $[f'(x)$ changes $+ \to -]$

  At $x = 1$: $f_{\min} = f(1) = 3(1) - 5(1) + 2 = 0$ $[f'(x)$ changes $- \to +]$

- **Concavity & Inflection Points:**

  $f''(x) = 60x^3 - 30x = 30x(2x^2 - 1) = 0 \implies x = -\frac{1}{\sqrt{2}},\ 0,\ \frac{1}{\sqrt{2}}$

  Concave Up $(f'' > 0)$: $\left(-\frac{1}{\sqrt{2}}, 0\right) \cup \left(\frac{1}{\sqrt{2}}, \infty\right),$

  Concave Down $(f'' < 0)$: $\left(-\infty, -\frac{1}{\sqrt{2}}\right) \cup \left(0, \frac{1}{\sqrt{2}}\right),$

  Inflection Points: $x = -\frac{1}{\sqrt{2}},\ 0,\ \frac{1}{\sqrt{2}}.$

Additional PUST EECE Past Exam Extrema Problems (Solved Vertically):

1. **PUST EECE 2022–23 Q2(c) [4 Marks]:** $f(x) = x^3 - 6x^2 + 9x + 1.$

   $f'(x) = 3x^2 - 12x + 9 = 3(x-1)(x-3) = 0 \implies x = 1,\ 3$

   $f''(x) = 6x - 12$

   At $x = 1$: $f''(1) = -6 < 0 \implies$ Local Maximum $f(1) = 5$

   At $x = 3$: $f''(3) = +6 > 0 \implies$ Local Minimum $f(3) = 1.$

2. **PUST EECE 2021–22 Q3(c) [4.5 Marks]:** $f(x) = 2x^3 - 21x^2 + 36x - 20.$

   $f'(x) = 6x^2 - 42x + 36 = 6(x-1)(x-6) = 0 \implies x = 1,\ 6$

   $f''(x) = 12x - 42$

   At $x = 1$: $f''(1) = -30 < 0 \implies$ Maximum value $f(1) = -3$

   At $x = 6$: $f''(6) = +30 > 0 \implies$ Minimum value $f(6) = -128.$

3. **PUST EECE 2020–21 Q4(a) [5 Marks]:** $f(x) = x^4 - 4x^3 + 10$.

   $f'(x) = 4x^3 - 12x^2 = 4x^2(x-3) = 0 \implies x = 0,\ 3$
   Decreasing on $(-\infty, 3]$, Increasing on $[3, \infty)$, Local Minimum $f(3) = -17$
   $f''(x) = 12x(x-2) = 0 \implies$ Inflection points at $(0, 10)$ and $(2, -6)$.

4.2 Working Rule for Curve Tracing

1. **Symmetry:** Even powers of $y \implies$ symmetric about x-axis; even powers of $x \implies$ symmetric about y-axis; unchanged under $x \leftrightarrow y \implies$ symmetric about $y = x$.
2. **Origin & Tangents at Origin:** If constant term is 0, curve passes through $(0, 0)$; equate lowest-degree terms to zero to find tangents at $(0, 0)$.
3. **Intercepts & Asymptotes:** Find x- and y-intercepts and all vertical, horizontal, and oblique asymptotes.
4. **Region of Exclusion & Derivatives:** Identify intervals where $y^2 < 0$ (no curve) and locate extrema/inflection points.

4.3 Past-Question Map for Module 3

Session	Questions Asked from Module 3 (Units 6–9)	Marks
2022–23	Q2(a) State & prove Rolle's theorem · Q2(c) Extrema of $x^3 - 6x^2 + 9x + 1$ · Q4(a) Maclaurin of $e^x \sec x$ · Q4(b) L'Hôpital limits	$5.5 + 4 + 4 + 5$
2021–22	Q1(b) Prove Lagrange's MVT · Q1(c) Verify Rolle for $x(x-3)^2$ · Q2(c) $3x^5 - 5x^3 + 2$ · Q3(c) Extrema · Q4(a,b) Maclaurin & L'Hôpital	$5 + 4 + 4 + 4.5 + 9.5$
2020–21	Q2(b) Verify MVT for $x(x-1)(x-2)$ on $[0, 1/2]$ · Q2(c) L'Hôpital limits · Q4(a) Extrema & concavity of $x^4 - 4x^3 + 10$	$4.5 + 4.5 + 5$

Quick Revision — Module 3 (Units 6–9)

• **Rolle's vs. MVT:** Rolle ($f(a) = f(b) \implies f'(c) = 0$); Lagrange MVT ($f'(c) = \frac{f(b)-f(a)}{b-a}$). Always verify $c \in (a, b)$ strictly inside the open interval!
• **Maclaurin Series:** $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$; $\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots$; $e^x \sec x = 1 + x + x^2 + \frac{2}{3}x^3 + \cdots$.
• **Extrema & Concavity:** $f'(c) = 0, f''(c) < 0 \implies$ Max; $f''(c) > 0 \implies$ Min; $f''(c) = 0 \implies$ use 1st Derivative sign chart!

Common Mistakes to Avoid in the Exam

• Using the Second Derivative Test when $f''(c) = 0$ (e.g., at $x = 0$ for $f(x) = 3x^5 - 5x^3 + 2$): $f''(c) = 0$ is inconclusive, so always show the $f'(x)$ sign table!
• Forgetting to take e^L at the end of an exponential indeterminate form ($1^\infty, 0^0, \infty^0$) after finding $L = \lim \ln y$.